Formler og figurer

- Til kurset E4IRT - Introduktion til reguleringsteknik

Dette dokument indeholder formler og figurer ofte anvendt i kurset E4IRT. Figurer og grafik er delvist lavet af underviser, delvist taget fra kursuslitteraturen (se kursets Brightspace-side for detaljer).

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Martin Ansbjerg Kjær, 2024-05-31
Aarhus Universitet, ECE

Elektriske kredsløb

Kirchoffs spændingslov (KVL): summen af spændingen i en lukket sløjfe er 0 V

Kirchoffs strømlov (KCL): summen af strømme ind (og ud) af et knudepunkt er 0 A

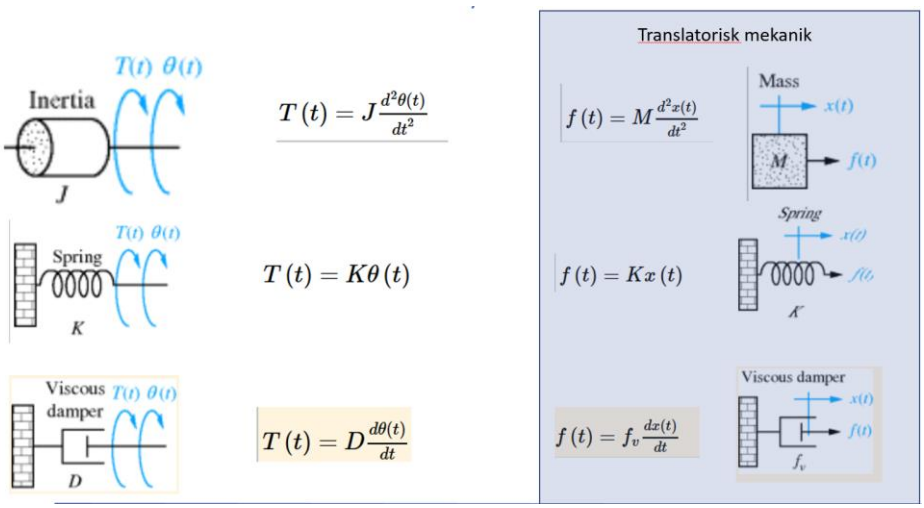
Impedansen af en modstand er $Z_R(s) = R$, hvor R er modstandens størrelse i Ohm

Impedansen af en spole er $Z_L(s) = Ls$, hvor L er spolens induktans i Henry (H)

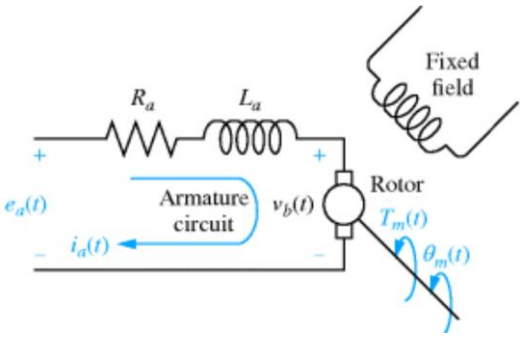
Impedansen af en kondensator er $Z_C(s) = \frac{1}{Cs}$, hvor C er spolens kapacitet i Farad (F)

Spænding $U_x(s)$ over en komponent med impedansen $Z_x(s)$ er relateret til strømmen gennem komponenten $I_x(s)$ gennem $U_x(s) = Z_x(s)\, I_x(s)$.

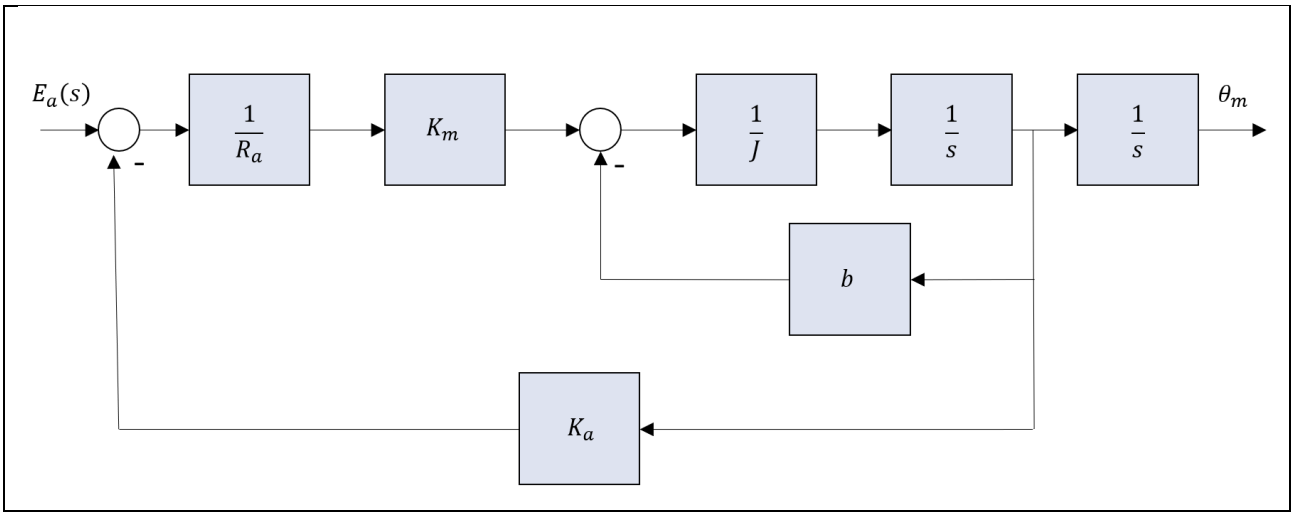
Mekaniske systemer



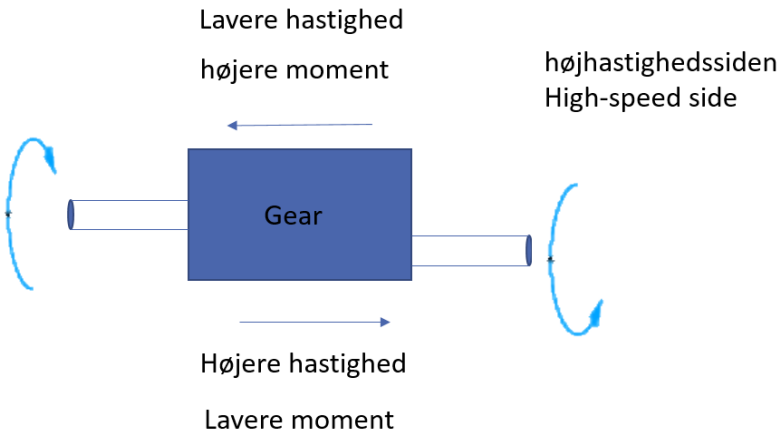
DC motoren



$$\frac{\theta(s)}{E_a(s)} = \frac{\frac{K_m}{JR_a}}{s\left[s + \frac{1}{J}\left(b + \frac{K_m K_a}{R_a}\right)\right]}$$



Gearing



Inertimomenter kan ”flyttes” på tværs af gear ved at

- dividere med n^2 når der flyttes fra lav- til højhastighedssiden
- gange med n^2 når der flyttes fra høj til lavhastighedssiden

Det samme gælder for ”D” og ”K”

Overføringsfunktioner

Generelle overføringsfunktioner kan beskrives ved deres n poler (p_i), m nulpunkter (z_i), og forstærkningsfaktor (K) ved

$$G(s) = \frac{Y(s)}{X(s)} = K\frac{(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$$

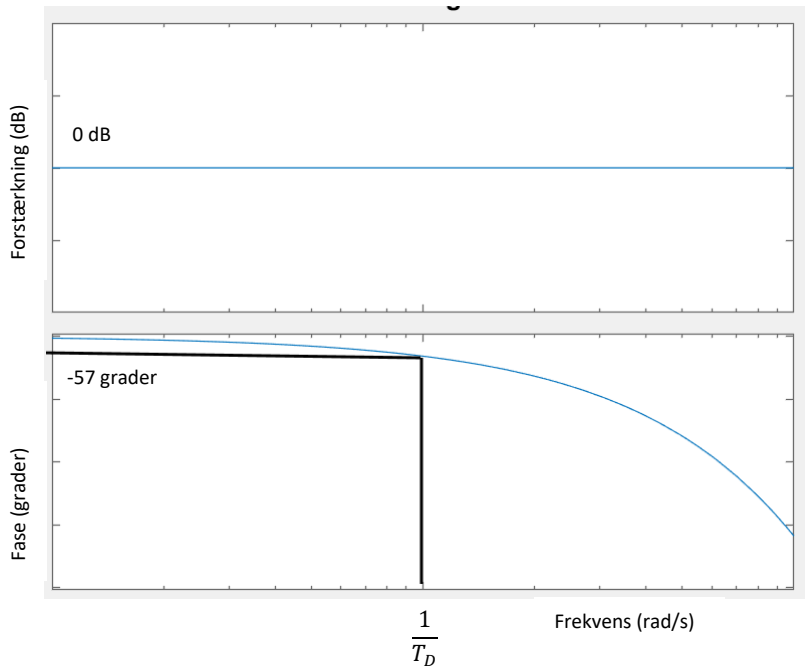
For et 1. ordenssystem uden nulpunkter, skrives dette ofte på en af følgende måder

$$G(s) = \frac{Y(s)}{X(s)} = \frac{K}{s+\alpha}$$

$$G(s) = \frac{Y(s)}{X(s)} = \frac{K\alpha}{s+\alpha}$$

$$G(s) = \frac{Y(s)}{X(s)} = \frac{K}{sT+1}$$

Forsinkelse

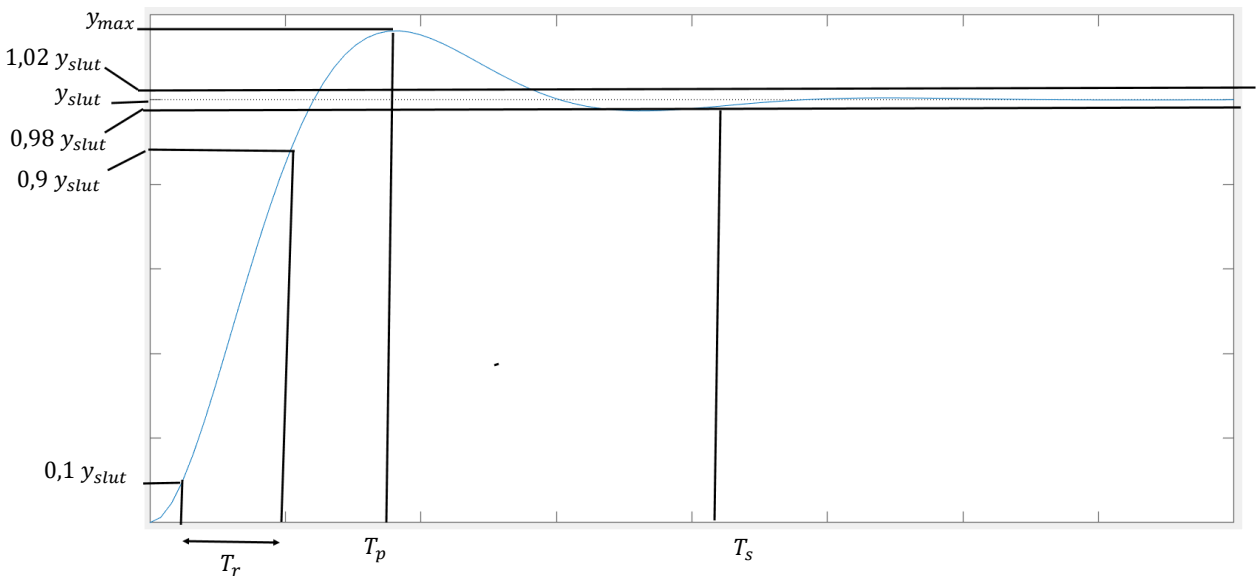


$$y(t) = x(t - T_D)$$

$$G_D(s) = \frac{Y(s)}{X(s)} = e^{-sT_D}$$

Analyse

Stepsvarsspecifikationer



Figur 1: Stepsvarsspecifikationer for et underdæmpet 2. ordens system uden nulpunkter.

Stationære fejl

For et system beskrevet ved åbensløjfeoverføringsfunktionen G(s), som reguleres med en enhedstilbagekobling gælder for den stationære kredsforstærkning K_0, den stationære forstærkning K_{stat}, og den stationære fejl $e(\infty)$ nedenstående tabel for relation mellem åben- og lukketsløjfe.

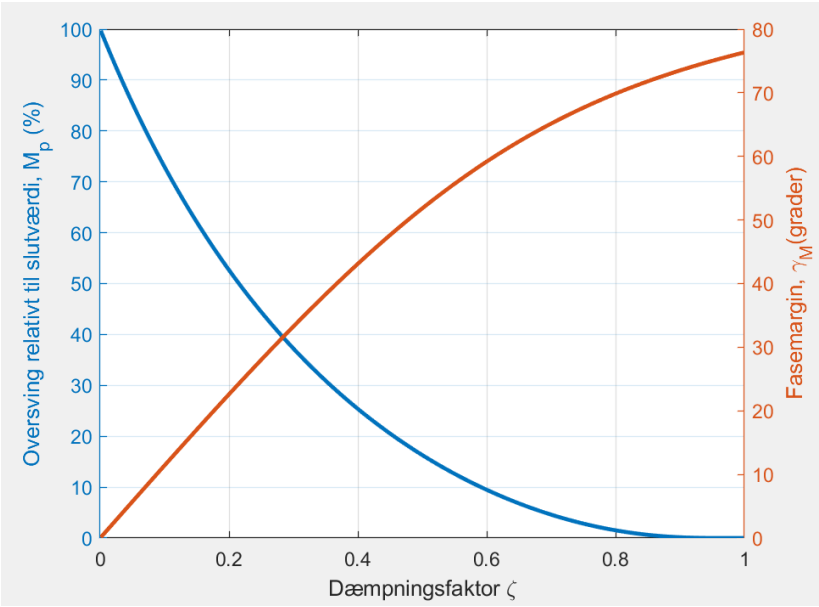
Type	K_0 $\left[\lim_{s\to 0} s^N G(s)\right]$	K_{stat} $\left[\lim_{s\to 0} G(s)\right]$	e(∞) ved step	e(∞) ved rampe
0	$\lim_{s\to 0} G(s)$	Konstant	$\frac{A}{1+K_0}$	$(\rightarrow)\ \infty$
1	$\lim_{s\to 0} sG(s)$	$(\rightarrow)\ \infty$	$(\rightarrow)\ 0$	$\frac{A}{K_0}$
2	$\lim_{s\to 0} s^2G(s)$	$(\rightarrow)\ \infty$	$(\rightarrow)\ 0$	$(\rightarrow)\ 0$
3	$\lim_{s\to 0} s^3G(s)$	$(\rightarrow)\ \infty$	$(\rightarrow)\ 0$	$(\rightarrow)\ 0$

Fasemargin og Oversving

For et lineært 2. ordens system med to poler og uden nulpunkter vil fasemargen Υ_m og oversvinget ved steps (%OS) være relateret til dæmpningsfaktoren ζ som angivet i Figur 2 og ved formlerne nedenfor.

$$G(S) = \frac{{\omega_n}^2 K_n}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$$

Hvor K_n er en frekvens-uafhængig forstærkningsfaktor.



Figur 2: Relation mellem oversving ved steps, dæmpningsfaktor ζ og fasemargin γ_M.

Den udæmpede egenfrekvens

Den *udæmpede egenfrekvens* ω_n for et 2. ordens lukketsløjfesystem uden nulpunkter vil tilnærmelsesvis være givet ved krydsfrekvensen ω_c.

Indsvingningstid

Indsvingningstiden (Settling time) t_s for et 2. ordens lukketsløjfesystem uden nulpunkter vil være relateret til den tilhørende ω_n og dæmpningsfaktor ζ, og kan tilnærmes med

- 2%-indsvingningstiden $t_{s,2} \cong \frac{4}{\zeta\,\omega_n}$
- 5%-indsvingningstiden $t_{s,5} \cong \frac{3}{\zeta\,\omega_n}$

Toptid

Toptid (Peak time) t_p for et 2. ordens lukketsløjfesystem uden nulpunkter vil være relateret til den tilhørende ω_n og dæmpningsfaktor ζ, og kan tilnærmes med

- Toptid $t_p \cong \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

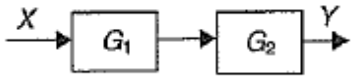
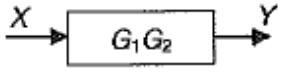
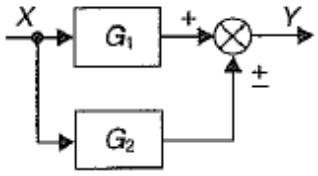
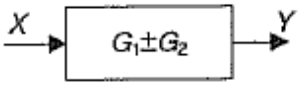
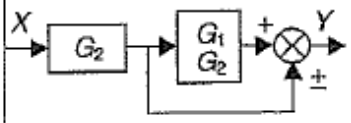
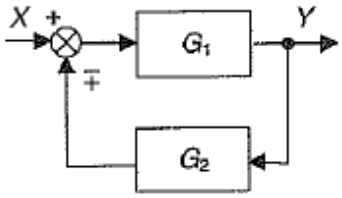
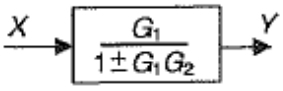
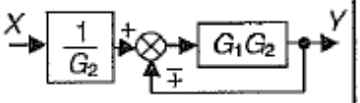
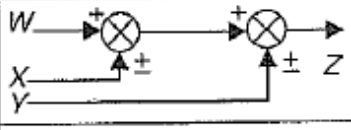
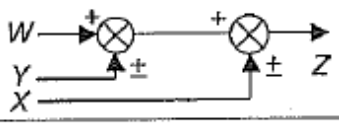
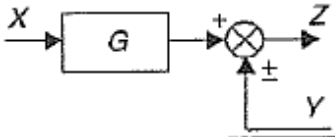
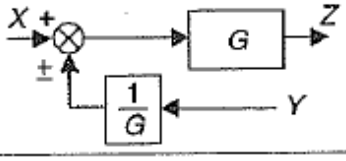
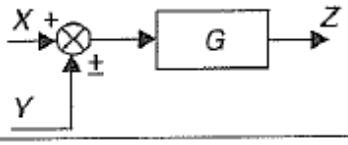
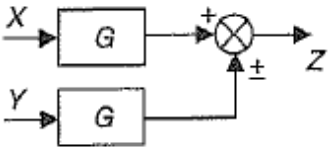
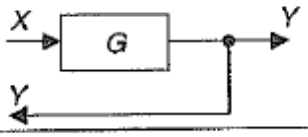
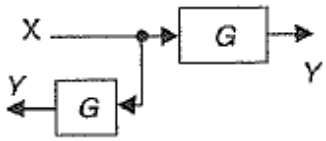
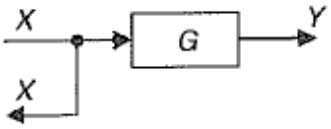
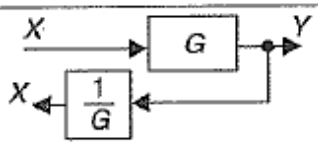
Stigtid

Stigtid (Rise time) t_r for et 2. ordens lukketsløjfesystem uden nulpunkter vil være relateret til den tilhørende ω_n og dæmpningsfaktor ζ, og kan tilnærmes med

- Stigtiden $t_r \cong \frac{1.8}{\omega_n}$

Blokdiagrammer

Table 4.1 Block Diagram Transformation Theorems

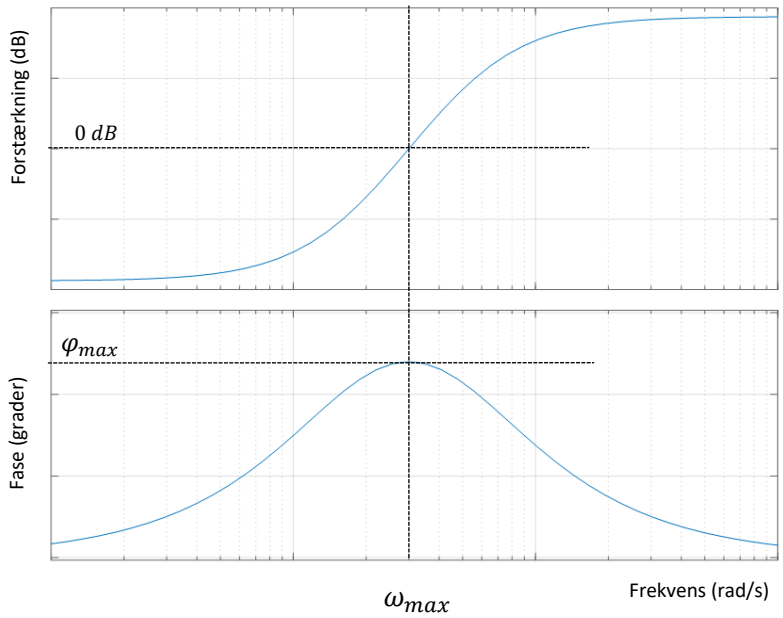
Transformation	Equation	Block diagram	Equivalent block diagram
1. Combining blocks in cascade	$Y=(G_1G_2)X$		
2. Combining blocks in parallel; or eliminating a forward loop	$Y=G_1X\pm G_2X$		
3. Removing a block from a forward path	$Y=G_1X\pm G_2X$		
4. Eliminating a feedback loop	$Y=G_1(X\pm G_2Y)$		
5. Removing a block from a feedback loop	$Y=G_1(X\pm G_2Y)$		
6. Rearranging summing points	$Z=W\pm X\pm Y$		
7. Moving a summing point ahead of a block	$Z=GX\pm Y$		
8. Moving a summing point beyond a block	$Z=G(X\pm Y)$		
9. Moving a take-off point ahead of a block	$Y=GX$		
10. Moving a take-off point beyond a block	$Y=GX$		

Regulatorer

Proportionalregulatoren

Proportionalregulatoren hæver og sænker forstærkningskurven i Bodeplottet. Fasen ændres ikke.

Lead regulatoren



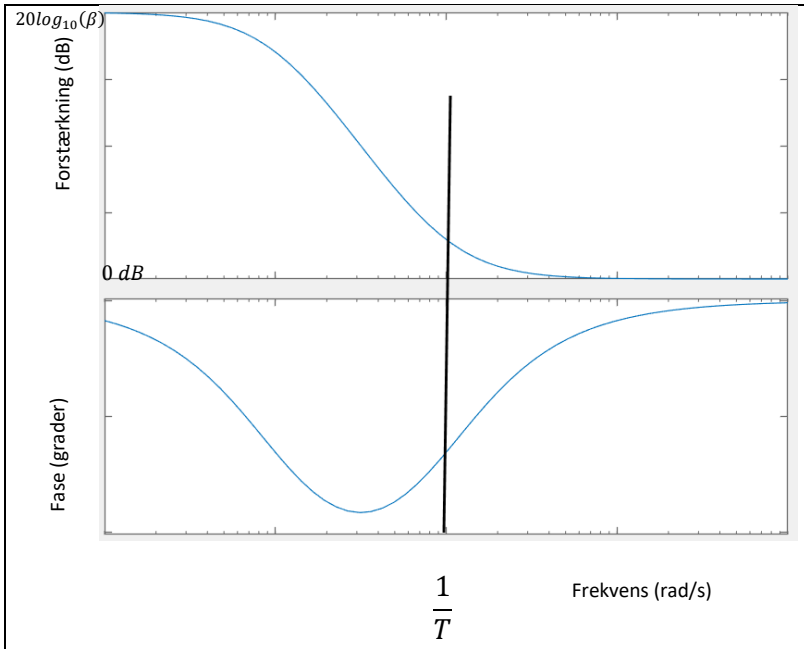
$$G_{\text{lead}}(s) = \frac{1}{\alpha}\frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}}K_c$$

$$\omega_{max} = \frac{1}{T\sqrt{\alpha}}$$

$$\alpha = \frac{1-\sin\varphi_{max}}{1+\sin\varphi_{max}}$$

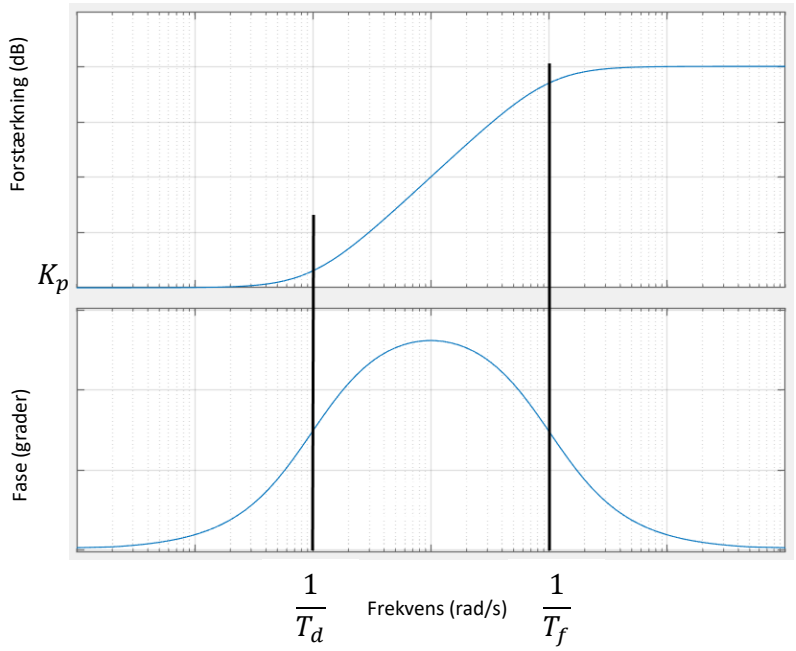
$$K_c = \sqrt{\alpha}$$

Lag regulatoren



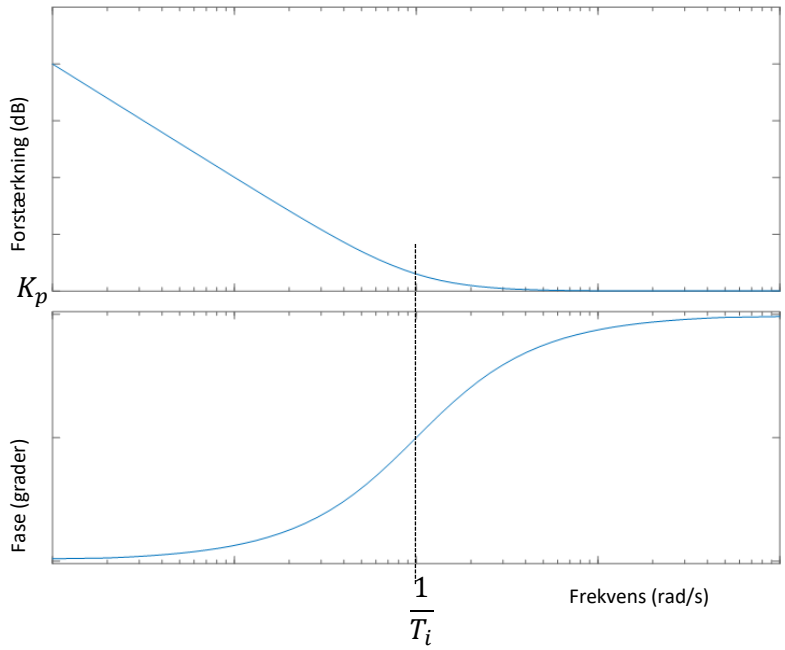
$$G_{\text{lag}}(s) = \frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}}$$

PD regulatoren



$$G_{\mathrm{PD}}(s) = K_p\left(1 + \frac{sT_d}{sT_f + 1}\right)$$

PI regulatoren



$$G_{\mathrm{PI}}(s) = K_p\left(1 + \frac{1}{sT_i}\right)$$

Digitalisering

Bilineære transformation

$$s = \frac{2}{T_s}\left(\frac{z-1}{z+1}\right)$$

ZOH

$$G_{p,zoh}(z) = (1 - z^{-1})Z\left\{\frac{G_p(s)}{s}\right\}$$

Samplingsfrekvens

Åström og Wittenmarks anbefaling til samplingsfrekvens

$$\frac{0{,}15}{\omega_c} \leq T_s \leq \frac{0{,}5}{\omega_c}$$

Bogen foreslår:

Vælg samplingsinterval 10 gange mindre end den dominerende tidskonstant

$$T_s \leq \frac{2\,\pi}{10\,\omega_{bw}} \approx \frac{2\,\pi}{10\,\omega_c}$$

Fasemargentab ved sampling

$$\varphi_{SH_loss} \approx \frac{\omega_c T_s}{2}$$

Z-transformationstabel

Table of Laplace and Z-transforms

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$
1.	–	–	Kronecker delta $\delta_0(k)$ 1 $k=0$ 0 $k \neq 0$	1
2.	–	–	$\delta_0(n-k)$ 1 $n=k$ 0 $n \neq k$	z^{-k}
3.	$\frac{1}{s}$	$1(t)$	$1(k)$	$\frac{1}{1-z^{-1}}$
4.	$\frac{1}{s+a}$	e^{-at}	e^{-akT}	$\frac{1}{1-e^{-aT}z^{-1}}$
5.	$\frac{1}{s^2}$	t	kT	$\frac{Tz^{-1}}{(1-z^{-1})^2}$
6.	$\frac{2}{s^3}$	t^2	$(kT)^2$	$\frac{T^2z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$
7.	$\frac{6}{s^4}$	t^3	$(kT)^3$	$\frac{T^3z^{-1}(1+4z^{-1}+z^{-2})}{(1-z^{-1})^4}$
8.	$\frac{a}{s(s+a)}$	$1-e^{-at}$	$1-e^{-akT}$	$\frac{(1-e^{-aT})z^{-1}}{(1-z^{-1})(1-e^{-aT}z^{-1})}$

Figur 3: Z-transformationstabel (https://www3.diism.unisi.it/~control/sdc/altro/TabellaTrasformataZ.pdf)

Matlab

Matlab komando	Eksempler	Matlab dokumentation
tf	s=tf('s') G=tf([1],[1 1])	Construct transfer function or convert to transfer function. SYS = tf(NUM,DEN) creates a continuous-time transfer function SYS with numerator NUM and denominator DEN. SYS is an object of type tf when NUM,DEN are numeric arrays, of type GENSS when NUM,DEN depend on tunable parameters (see REALP and GENMAT), and of type USS when NUM,DEN are uncertain (requires Robust Control Toolbox). SYS = tf(NUM,DEN,TS) creates a discrete-time transfer function with sample time TS (set TS=-1 if the sample time is undetermined). S = tf('s') specifies the transfer function H(s) = s (Laplace variable). Z = tf('z',TS) specifies H(z) = z with sample time TS.
bode	bode(G) bode(G*Gc)	Bode frequency response of dynamic systems. bode(SYS) draws the Bode plot of the dynamic system SYS.

margin	margin(G) margin(G*Gc)	Gain and phase margins and crossover frequencies. [Gm,Pm,Wcg,Wcp] = margin(SYS) computes the gain margin Gm, the phase margin Pm, and the associated frequencies Wcg and Wcp,
feedback	Gcl=feedback(G*Gpi,1)	Feedback connection of two input/output systems. M = feedback(M1,M2) computes a closed-loop model M for the feedback loop: u --->O---->[M1]----+---> y | | y = M * u +-----[M2]<---+ Negative feedback is assumed and the model M maps u to y.
c2d	c2d(G,0.1,'zoh')	c2d Converts continuous-time dynamic system to discrete time. SYSD = c2d(SYSC,TS,METHOD) computes a discrete-time model SYSD with sample time TS that approximates the continuous-time model SYSC. The string METHOD selects the discretization method among the following: 'zoh' Zero-order hold on the inputs 'foh' Linear interpolation of inputs 'impulse' Impulse-invariant discretization 'tustin' Bilinear (Tustin) approximation. 'matched' Matched pole-zero method (for SISO systems only). The default is 'zoh' when METHOD is omitted. The sample time TS should be specified in the time units of SYSC (see "TimeUnit" property).
step	step(Gcl)	Step response of dynamic systems. [Y,T] = step(SYS) computes the step response Y of the dynamic system SYS.